

Application of U-net neural network for automatic spike detection in EEG recordings from patients with rolandic epilepsy

Albina Lebedeva

Laboratory of AI and Bid Data

Lobachevsky University

Privolzhsky Research Medical University

Nizhny Novgorod, Russia

lebedeva@neuro.nnov.ru

Margarita Samburova

Laboratory of AI and Bid Data

Lobachevsky University

Nizhny Novgorod, Russia

samburova@unn.ru

Oksana Elshina

Privolzhsky Research Medical University

Nizhny Novgorod, Russia

elshinaod@mail.ru

Anastasiya Borisova

Privolzhsky Research Medical University

Nizhny Novgorod, Russia

anastasia.gostenko@gmail.com

Valentin Borisov

Privolzhsky Research Medical University

Nizhny Novgorod, Russia

valyaborisov@gmail.com

Nikolay Gromov

Laboratory of AI and Bid Data

Lobachevsky University

Nizhny Novgorod, Russia

gromov@itmm.unn.ru

Artem Sharkov

Laboratory of AI and Bid Data

Lobachevsky University

Nizhny Novgorod, Russia

Veltishev Research and Clinical Institute for Pediatrics and Pediatric Surgery

Pirogov Russian National Research Medical University

Moscow, Russia

a.a.sharkov@yandex.ru

Maria Yashanova

Privolzhsky Research Medical University

Nizhny Novgorod, Russia

yammi2006@gmail.com

Svetlana Gerasimova

Scientific and Educational Mathematical Center

Lobachevsky University

Nizhny Novgorod, Russia

gerasimova@neuro.nnov.ru

Tatiana Levanova*

Laboratory of AI and Bid Data

Lobachevsky University

Nizhny Novgorod, Russia

tatiana.levanova@itmm.unn.ru

Lev Smirnov

Laboratory of AI and Bid Data

Lobachevsky University

Nizhny Novgorod, Russia

lev.smirnov@itmm.unn.ru

Abstract—The goal of the proposed study is to implement and test AI approaches to segmentation of benign epilepsy with centro-temporal spikes (BECTS) patterns in electroencephalogram (EEG) recordings. In order to do this we apply U-net-type deep neural network previously developed for electrocardiogram (ECG) segmentation. Our results shows potential applicability of this approach and can be used as a basis for further research in the field.

Index Terms—rolandic epilepsy, rolandic spikes, deep learning

I. INTRODUCTION

Rolandic epilepsy is the most common form of epileptic encephalopathy, characterized by sleep-induced inferior rolandic epileptiform spikes, seizures, and cognitive impairment in school-age children. This condition can spontaneously

be resolved by adolescence. The primary focus of Rolandic spikes is the Rolandic (central) sulcus. The central sulcus (Latin: sulcus centralis cerebri), also known as the Rolandic sulcus (Latin: sulcus Rolandi, after the Italian anatomist Luigi Rolando who described it), is a large groove in the cerebral cortex separating the frontal lobe from the parietal lobe [1]. Recently, it has been shown that many of these children have a high risk of developing cognitive impairment. Moreover, abnormal EEG activity persists long after remission.

As with other epilepsy syndromes, the diagnosis of benign epilepsy with centro-temporal spikes (BECTS) is based on history and should be confirmed by characteristic electroencephalogram (EEG) findings [2]. In this case EEG shows slow, biphasic (negative discharges in the central temporal area and positive discharges in the frontal area), high-voltage, central temporal sharp spikes, often followed by a slow wave. These sharp spikes occur in repetitive bursts, and are usually

The study was supported by the Ministry of Science and Higher Education of Russian Federation (Contract No. FSWR-2024-0005).

unilateral (consistent with the focal nature of the seizures) but may occasionally be bilateral. The EEG should always include a wakefulness and sleep recording. Sleep recording is very important because spikes are sometimes only visible during sleep. Non-rapid eye movement (non-REM) sleep recordings are of greatest value because they highlight epileptiform activity [3]. However, it often happens that the epileptiform activity characteristic of rolandic epilepsy can transform into another type of epilepsy, that is, the disease can follow a negative scenario. The mechanisms of such a scenario are currently unknown and require further study [4]. In addition, accurate diagnosis by a doctor also requires the development of new automated methods, e.g., those including viewing and counting the number of epileptiform graphoelements in EEG records.

In clinical practice, the relationship between the location and duration of epileptiform discharges of the Rolandic area in the entire EEG monitoring is widely used as the main indicators for diagnosing patients with BECTS and determining the need for medication [5]. This indicator has become an important criterion for determining whether BECTS patients need drug therapy. Therefore, the detection of spikes can help doctors make more accurate diagnoses and prescribe more accurate treatments for patients with BECTS.

In practice, EEG specialists usually read EEG recordings and identify BECTS by manual recognition. However, the detection results often vary due to the subjectivity of different specialists. With the increase of EEG recordings, misjudgments caused by visual fatigue and other factors will inevitably occur. Therefore, the development and application of automatic epileptic spike detection algorithms can effectively improve the accuracy and efficiency of diagnosis.

Artificial intelligence technologies can be most effective in this area [6]. Machine learning and deep learning algorithms are widely used for BECTS-related tasks, see, e.g. [6]–[9]. In the basis of sufficient parts of proposed architectures lies the automatic feature extraction, which can be performed using convolutional layers. Therefore, convolutional neural networks can be viewed as the first choice deep architectures for BECTS detection and segmentation [10]. However, the implementation of these results in clinical practice in the field of neurology is yet to come, which emphasizes the need to assess the problems and limitations of machine learning and deep learning algorithms [11].

The goal of the proposed study is to implement and test AI approaches to segmentation of BECTS patterns in EEG recordings. In order to do this we apply U-net-type deep neural network previously developed for electrocardiogram (ECG) segmentation for the segmentation of BECTS in EEG. Our results shows potential applicability of this approach and can be used as a basis for further research in the field.

II. METHODS

A. Ethics Statement

In this study, the neural network training protocols using anonymized EEG data were reviewed and approved by the Bioethics Committee of the Institute of Biology and

Biomedicine of the National Research Lobachevsky State University of Nizhny Novgorod.

B. EEG data collection

All EEG records are presented in digital format (EDF, EEG-ASCII, UDF). Recordings from patients made in a calm waking state were anonymized and transferred to us for subsequent analysis from the University Clinic of PIMU. All EEG records were made with electrode leads according to the "10-20" system (21 leads) with an additional ECG lead. The duration of the records taken into analysis ranged from 10 minutes to 2 hours.

C. Deep learning architecture

The modified U-Net architecture [12] was implemented for segmentation of BECTS patterns in the EEG recordings. The schematic representation of this deep learning architecture is presented in Fig. 1. The choice of this neural network architecture was determined by the fact of visual similarity of rolandic spikes and QRS complexes in the heart rhythm in ECG recordings.

This deep learning network takes one lead (single time series) as an input for ECG segmentation. One of the leads in the 10 – 20 electrode lead system, namely, F7-T3, was selected for the further consideration. This lead was chosen for consideration due to its maximum proximity to the Rolandic sulcus, which means that in this lead BECTS patterns will be observed in greater quantities and will be easily recognizable.

D. Preprocessing and training

All records used in the study were preliminarily manually segmented by qualified epileptologists. Then obtained data were split into train and test datasets with respect to the separation by patients. This ensures that training process is performed in such a way that the network does not see any EEG examples of patients from the test sample during the training phase. After that the resulting 10-minute records were divided into 10-second segments. Since the sampling frequency of the records was 500 Hz, each item (segment) contains 5000 points. The manual markup performed by epileptologists was also converted into a mask of length 5000, where 0 meant the absence of a rolandic spike, and 1 meant its presence. Thus, the obtained dataset can be presented in the following form:

$$\begin{aligned} X &= (x_{ij}), \\ Y &= (y_{ij}), y_{ij} \in \{0, 1\}, \\ i &= 1, \dots, 233, \\ j &= 1, \dots, 5000. \end{aligned}$$

The deep learning neural network was trained using cross-validation approach over 70 epochs, with cross entropy as the loss function and Adam optimizer.

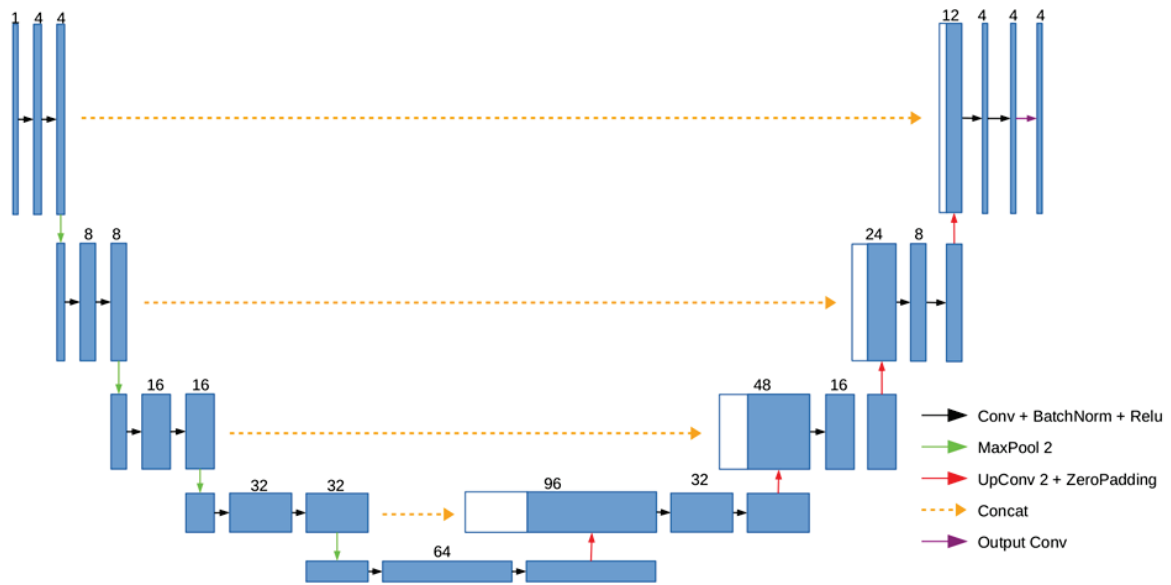


Fig. 1. The schematic representation of deep learning architecture (adapted from [12]).

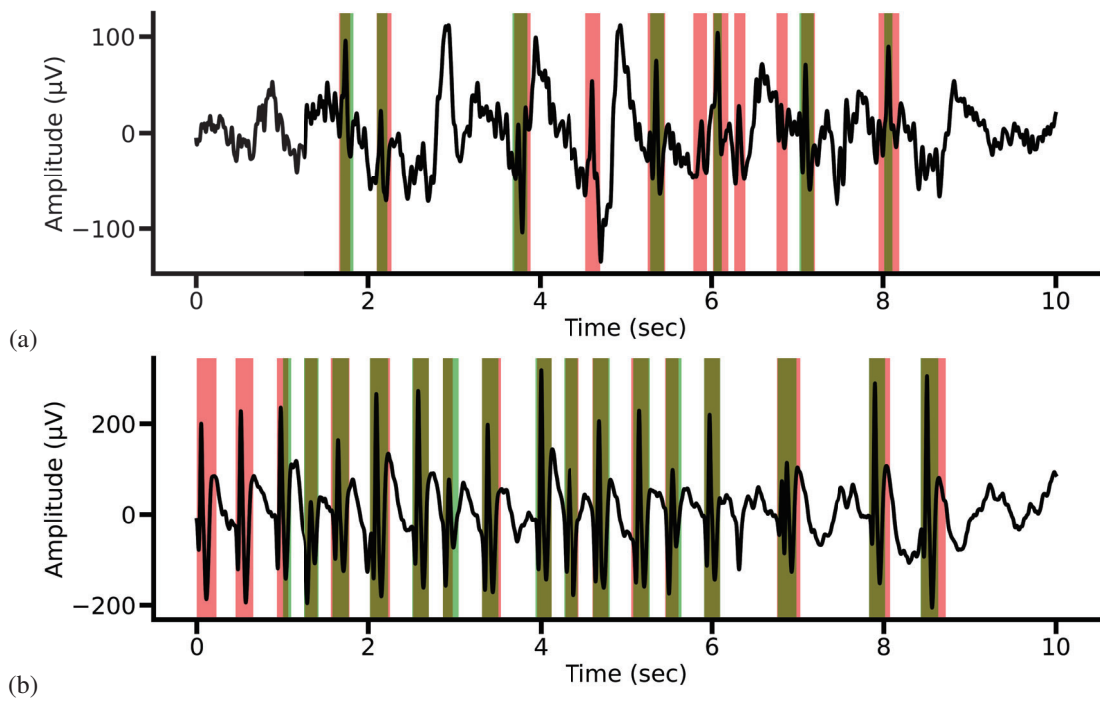


Fig. 2. Segmented BECTs in EEG recording. Here red marks are set manually by epileptologists, green marks are set automatically by deep neural network. (a) Too many unlabeled BECT patterns in the recording. (b) Correct labelling of BECT patterns almost in all cases.

E. Evaluation metrics

The dice coefficient (DC) was calculated in order to estimate the difference between manual segmentation performed by epileptologists and automatic segmentation performed by chosen deep learning architecture for each cross-validation iteration and for each EEG instance. Mathematically it can be written as follows:

$$DC = \frac{2|X \cap Y|}{|X| + |Y|},$$

where X are the boundaries of the labeling performed by epileptologists, and Y are the boundaries of the labeling performed by deep neural network.

III. RESULTS

It was shown that the average value of DC for the manual and automatic segmentations was 0.62, which means that rolandic spikes were successfully detected in half of the cases (at a significance level of 0.05, with p-value 0.177).

Examples of different segmented EEG recordings can be seen in Fig. 2. As can be seen from Fig. 2, this U-net-type deep learning network is able to capture the valuable morphological features of BECTS quite well in general. Typically, unlabeled spikes refer to the beginning and end of a recording. Also, the spike boundaries are not always determined correctly, e.g., the network sets the pattern boundaries narrower than they are marked by the doctor. Such cases also worsen the evaluation metric value, despite the fact that the spike itself is determined correctly in general.

These problems may be a consequences of insufficient fitting caused by a small training set. Since the proposed U-net-type deep learning network is a convolutional architecture, it contains many trainable weights (adjustable parameters), which should be optimized during training process. Thus, the described problems can presumably be eliminated by increasing the number of EEG recordings from different patients in the training set.

IV. CONCLUSION

This paper presents an original approach to automatic detection of rolandic spikes (BECTS) in anonymized EEG records from patients with a confirmed diagnosis of rolandic epilepsy. The approach is based on the use of a modified U-net deep neural network designed to detect QRS complexes in ECG records. At this stage, the proposed approach is implemented for recording EEG from single lead (which is chosen as the lead with the maximum amplitude of rolandic spikes). It was found that the proposed deep neural network is capable of automatically detecting rolandic spikes in general. The obtained value of DC evaluation metric was 50%, which is quite good for such a small data set. Our results shows that described architecture is potentially interesting for BECT segmentation task and can achieve better values of evaluation metric with further improvements.

First, this result presumably can be improved by including more EEG recordings manually labeled by epileptologists in

the training set. Next, we plan modify the architecture of deep learning network in the way that will allow it to process all leads in the EEG recordings simultaneously and therefore advance from correlations between leads.

This study is the first step in developing of a custom software product for BECTS segmentation purposes. We hope that our findings in future will be effective and useful in the clinical practice in the field of epileptology and will contribute to improving the quality of diagnoses.

REFERENCES

- [1] R. Nieuwenhuys, J. Voogd, and C. Van Huijzen, *The human central nervous system: a synopsis and atlas*. Springer Science & Business Media, 2007.
- [2] A. Datta and D. B. Sinclair, "Benign epilepsy of childhood with rolandic spikes: typical and atypical variants," *Pediatric neurology*, vol. 36, no. 3, pp. 141–145, 2007.
- [3] R. Samaitienė, J. Norkūnienė, B. Tumienė, and J. Grikinienė, "Sleep and behavioral problems in rolandic epilepsy," *Pediatric neurology*, vol. 48, no. 2, pp. 115–122, 2013.
- [4] C. Garcia-Ramos, D. C. Jackson, J. J. Lin, K. Dabbs, J. E. Jones, D. A. Hsu, C. E. Stafstrom, L. Zawadzki, M. Seidenberg, V. Prabhakaran *et al.*, "Cognition and brain development in children with benign epilepsy with centrottemporal spikes," *Epilepsia*, vol. 56, no. 10, pp. 1615–1622, 2015.
- [5] Z. Xu, T. Wang, J. Cao, Z. Bao, T. Jiang, and F. Gao, "Bect spike detection based on novel eeg sequence features and lstm algorithms," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 1734–1743, 2021.
- [6] D. Nhu, M. Janmohamed, A. Antonic-Baker, P. Perucca, T. J. O'Brien, A. K. Gilligan, P. Kwan, C. W. Tan, and L. Kuhlmann, "Deep learning for automated epileptiform discharge detection from scalp eeg: A systematic review," *Journal of Neural Engineering*, vol. 19, no. 5, p. 051002, 2022.
- [7] L. Jiang, Q. Fan, J. Ren, F. Dong, T. Jiang, and J. Liu, "An improved bect spike detection method with functional brain network features based on plv," *Frontiers in Neuroscience*, vol. 17, p. 1150668, 2023.
- [8] A. Sathyanarayana, R. El Atrache, M. Jackson, A. S. Alter, K. D. Mandl, T. Loddenkemper, and W. J. Bosl, "Nonlinear analysis of visually normal eegs to differentiate benign childhood epilepsy with centrottemporal spikes (bects)," *Scientific Reports*, vol. 10, no. 1, p. 8419, 2020.
- [9] T. Luo, J. Wang, Y. Zhou, S. Zhou, C. Hu, P. Yao, Y. Zhang, and Y. Wang, "Emd-wog-2dcnn based eeg signal processing for rolandic seizure classification," *Computer Methods in Biomechanics and Biomedical Engineering*, vol. 25, no. 14, pp. 1565–1575, 2022.
- [10] J. Wu, Z. Xiong, X. He, D. Wu, T. Jiang, and L. Jiang, "Automatic bect spike detection algorithm based on adaptive template matching and convolution neural network," 2024.
- [11] A. Lucas, A. Revell, and K. A. Davis, "Artificial intelligence in epilepsy—applications and pathways to the clinic," *Nature Reviews Neurology*, pp. 1–18, 2024.
- [12] V. Moskalenko, N. Zolotikh, and G. Osipov, "Deep learning for eeg segmentation," in *Advances in Neural Computation, Machine Learning, and Cognitive Research III: Selected Papers from the XXI International Conference on Neuroinformatics, October 7-11, 2019, Dolgoprudny, Moscow Region, Russia*. Springer, 2020, pp. 246–254.